

Ontwikkelperslag G2

Opgemaakt 2026-10-05 11:29 CEST uit het generatieregister en het evolutieregister. Status:

survived

1. Selectie

Besluit vastgelegd in het evolutieregister (rij 16). Onderbouwing:

```
{
  "forgetting_P": 0.575,
  "forgetting_min": 0.25,
  "promotion": {
    "candidate_iqm": 0.8069688889671774,
    "p_value": 0.003968253968253968,
    "prob_improvement": 1.0,
    "promote": true,
    "reference_iqm": 0.46979370481231647
  },
  "rule": "decide_promotion(G2, G1 on the 9-task battery; G1 = 0 on new tasks) and P(G2 > G1 on G1 tasks) >= forgetting_min",
  "verdict": "survived"
}
```

2. Populatie

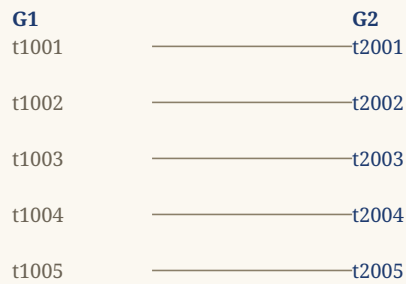
toddler	rol	registratie	geërfd van (modules)
G2/t2001	population	born	G1/t1001: trunk, task:cartpole, task:acrobot, task:empty5, task:doorkey5
G2/t2002	population	born	G1/t1002: trunk, task:cartpole, task:acrobot, task:empty5, task:doorkey5
G2/t2003	population	born	G1/t1003: trunk, task:cartpole, task:acrobot, task:empty5, task:doorkey5
G2/t2004	population	born	G1/t1004: trunk, task:cartpole, task:acrobot, task:empty5, task:doorkey5
G2/t2005	population	born	G1/t1005: trunk, task:cartpole, task:acrobot, task:empty5, task:doorkey5

3. Overerving (generatiedata)

kind	ouder	sha256 oudergewichten
G2/t2001	G1/t1001	eddd1a6bf5c41c851b9a136abcd4b7e4b2c6f273f44cf1e69d9d6c9e70bea4b
G2/t2002	G1/t1002	402e4cacb7e065deb67ed4941261829cfadaacf1f91e493c2a56466353fc5584
G2/t2003	G1/t1003	d6713faa21065fe4d68551721d2eae62079c40bf13f36cdeecb42bd287b97e4c
G2/t2004	G1/t1004	02396afd093d615d33de27f7ed07b8c4673ded187674b3c4a526cb797ed5c755
G2/t2005	G1/t1005	8bed186b8a76ad8dcdf348267dc1d53435002fc4a72ce9f99dd4c8896692c969

Elke geërfde ouder is bij de geboorte gecontroleerd tegen zijn eigen registratie; een afwijkende hash wordt door het register geweigerd.

4. Stamboom



5. Hardware

```
{
  "cores": 96,
  "cpu": "x86_64",
  "device": "cpu",
  "gpus": [
    "NVIDIA RTX 4000 Ada Generation",
    "NVIDIA RTX 4000 Ada Generation",
    "NVIDIA RTX 4000 Ada Generation",
    "Tesla V100-SXM2-32GB",
    "Tesla V100-SXM2-32GB",
    "NVIDIA RTX 4000 Ada Generation"
  ],
  "torch_threads": 4
}
```

6. Software en code

```
{
  "commit": "9e4dfc29a88b2ed215232c31555a54ad9577c4a6",
  "dirty": false,
  "remote": "git@github.com:virtuanalytica/toddler.git"
}

{
  "gymnasium": "1.2.1",
  "numpy": "2.3.1",
  "torch": "2.6.0+cu124"
}
```

7. Trainingsdata en evaluatie

```
{
  "env_versions": {
    "gymnasium": "1.2.1",
    "minigrid": "3.1.0"
  },
  "hidden_seed_set": {
    "commitment": "e2b94c7d6946d8cd53ec829016629acf31f9e0c3fd4129d3945d6871d966cfc4",
    "created_at": "2026-10-05T08:18:01+00:00",
    "n": 30,
    "reveal_after": "2026-10-12",
    "seed_high": 1000000,
    "seed_low": 100000,
    "set_id": "20261005T081801Z-0244f5"
  }
}
```

```

},
"public_eval_seeds_sha256": "79bb167f2fa0b45930420359382fc6bcf6a7571935a58f876d90b110a91ca1a",
"solve_thresholds": {
  "acrobot": -100.0,
  "cartpole": 475.0,
  "doorkey5": 0.9,
  "doorkey8": 0.9,
  "empty5": 0.9,
  "keycorridor3": 0.9,
  "lavacross9": 0.9,
  "unlock": 0.9,
  "unlockpickup": 0.9
},
"tasks": {
  "acrobot": "Acrobot-v1",
  "cartpole": "CartPole-v1",
  "doorkey5": "MiniGrid-DoorKey-5x5-v0",
  "doorkey8": "MiniGrid-DoorKey-8x8-v0",
  "empty5": "MiniGrid-Empty-5x5-v0",
  "keycorridor3": "MiniGrid-KeyCorridorS3R1-v0",
  "lavacross9": "MiniGrid-LavaCrossingS9N1-v0",
  "unlock": "MiniGrid-Unlock-v0",
  "unlockpickup": "MiniGrid-UnlockPickup-v0"
},
"training_seed_band": [
  1000000,
  2147483648
]
}

```

Een verborgen seed-set wordt alleen met zijn commitment vermeld tot het venster sluit; daarna is hij te controleren met `python3 -m toddler.learn.secret_seeds reveal <set_id>`.

8. Budget

toddler	budget
G2/ t2001	<pre> {"block_steps": 25000, "eval_mode": "sample", "seconds": 4202.4, "seed": 2001, "steps": {"acrobot": 150000, "cartpole": 150000, "doorkey5": 150000, "doorkey8": 1000000, "empty5": 150000, "keycorridor3": 1000000, "lavacross9": 1000000, "unlock": 1000000, "unlockpickup": 1000000}, "threads": 4} </pre>
G2/ t2002	<pre> {"block_steps": 25000, "eval_mode": "sample", "seconds": 4183.8, "seed": 2002, "steps": {"acrobot": 150000, "cartpole": 150000, "doorkey5": 150000, "doorkey8": 1000000, "empty5": 150000, "keycorridor3": 1000000, "lavacross9": 1000000, "unlock": 1000000, "unlockpickup": 1000000}, "threads": 4} </pre>
G2/ t2003	<pre> {"block_steps": 25000, "eval_mode": "sample", "seconds": 4053.1, "seed": 2003, "steps": {"acrobot": 150000, "cartpole": 150000, "doorkey5": 150000, "doorkey8": 1000000, "empty5": 150000, "keycorridor3": 1000000, "lavacross9": 1000000, "unlock": 1000000, "unlockpickup": 1000000}, "threads": 4} </pre>
G2/ t2004	<pre> {"block_steps": 25000, "eval_mode": "sample", "seconds": 4087.8, "seed": 2004, "steps": {"acrobot": 150000, "cartpole": 150000, "doorkey5": 150000, "doorkey8": 1000000, "empty5": 150000, "keycorridor3": 1000000, "lavacross9": 1000000, "unlock": 1000000, "unlockpickup": 1000000}, "threads": 4} </pre>
G2/ t2005	<pre> {"block_steps": 25000, "eval_mode": "sample", "seconds": 4043.5, "seed": 2005, "steps": {"acrobot": 150000, "cartpole": 150000, "doorkey5": 150000, "doorkey8": 1000000, "empty5": 150000, "keycorridor3": 1000000, "lavacross9": 1000000, "unlock": 1000000, "unlockpickup": 1000000}, "threads": 4} </pre>

9. Resultaten (generatierapport)

```
{
  "generation": "G2",
  "tasks": [
    "cartpole",
    "acrobot",
    "empty5",
    "doorkey5",
    "doorkey8",
    "unlock",
    "unlockpickup",
    "keycorridor3",
    "lavacross9"
  ],
  "new_tasks": [
    "doorkey8",
    "unlock",
    "unlockpickup",
    "keycorridor3",
    "lavacross9"
  ],
  "steps": {
    "cartpole": 150000,
    "acrobot": 150000,
    "empty5": 150000,
    "doorkey5": 150000,
    "doorkey8": 1000000,
    "unlock": 1000000,
    "unlockpickup": 1000000,
    "keycorridor3": 1000000,
    "lavacross9": 1000000
  },
  "block_steps": 25000,
  "eval_mode": "sample",
  "method_version": {
    "scale_rewards": true,
    "network": "multitask",
    "inherit": "G1"
  },
  "measured_at": "2026-10-05T11:28:26+0200",
  "training_seconds": 4224.7,
  "reference_fingerprint": "8e69e3b86f262131cb9ae4caa8e70080e353749bcc9829e0614136ad50eb7e4b",
  "g1_reference_fingerprint": "8978f072b40ca13fdf79db1f1b12c05286e52c967a6951a08fbaa02efc178663",
  "secret_seed_commitment": {
    "set_id": "20261005T081801Z-0244f5",
    "n": 30,
    "seed_low": 100000,
    "seed_high": 1000000,
    "created_at": "2026-10-05T08:18:01+00:00",
    "reveal_after": "2026-10-12",
    "commitment": "e2b94c7d6946d8cd53ec829016629acf31f9e0c3fd4129d3945d6871d966cfc4"
  },
  "arms": {
    "G2": {
      "per_task_iqm_public": {
        "cartpole": 1.0552,
        "acrobot": 1.0359,
        "empty5": 1.0723,
        "doorkey5": 1.0729,
        "doorkey8": 0.0352,
        "unlock": 1.0757,
        "unlockpickup": 0.0,
        "keycorridor3": 0.6795,
        "lavacross9": 0.8856
      },
      "per_task_iqm_secret": {
```

```

"cartpole": 1.0553,
"acrobot": 1.0436,
"empty5": 1.0723,
"doorkey5": 1.0733,
"doorkey8": 0.0029,
"unlock": 1.0779,
"unlockpickup": 0.0,
"keycorridor3": 0.6837,
"lavacross9": 0.9485
},
"new_tasks_iqm_public": [
  0.5352,
  [
    0.3874,
    0.7076
  ]
],
"new_tasks_iqm_secret": [
  0.5426,
  [
    0.3928,
    0.7094
  ]
],
"IQ_quotient_vs_G1_on_G1_tasks": [
  100.2,
  [
    99.7,
    100.6
  ]
],
"toddlers": [
  "t2001",
  "t2002",
  "t2003",
  "t2004",
  "t2005"
]
},
"G2-scratch": {
  "per_task_iqm_public": {
    "cartpole": 1.0552,
    "acrobot": 1.0239,
    "empty5": 1.0723,
    "doorkey5": 1.0704,
    "doorkey8": 0.7135,
    "unlock": 1.0702,
    "unlockpickup": 0.0,
    "keycorridor3": 1.0556,
    "lavacross9": 0.9095
  },
  "per_task_iqm_secret": {
    "cartpole": 1.0553,
    "acrobot": 1.0331,
    "empty5": 1.0723,
    "doorkey5": 1.0723,
    "doorkey8": 0.7041,
    "unlock": 1.0715,
    "unlockpickup": 0.0,
    "keycorridor3": 1.0551,
    "lavacross9": 0.9547
  },
  "new_tasks_iqm_public": [
    0.7498,
    [
      0.5845,
      0.8447
    ]
  ]
}

```

```

],
"new_tasks_iqm_secret": [
  0.7571,
  [
    0.5985,
    0.847
  ]
],
"IQ_quotient_vs_G1_on_G1_tasks": [
  99.8,
  [
    94.8,
    100.2
  ]
],
"toddlers": [
  "t2001",
  "t2002",
  "t2003",
  "t2004",
  "t2005"
]
},
"inheritance_effect": {
  "P(G2 > G2-scratch) public, new tasks": 0.436,
  "P(G2 > G2-scratch) secret, new tasks": 0.448,
  "P(G2 > G1) public, G1 tasks (forgetting check)": 0.575
},
"public_vs_secret_gap_new_tasks": {
  "G2": -0.0074,
  "G2-scratch": -0.0073
},
"IQ_note": "IQ quotient on the four G1 tasks against the frozen G1 reference; G2 is the frozen reference for the nine-tas",
"EQ_FQ": "not measured (no judged tasks or reflex scenarios yet)",
"invoicing_entity": "VirtualV Holding B.V.",
"compute_and_sales_entity": "virtuanalytica VOF (commission)",
"selection": {
  "verdict": "survived",
  "promotion": {
    "promote": true,
    "p_value": 0.003968253968253968,
    "prob_improvement": 1.0,
    "candidate_iqm": 0.8069688889671774,
    "reference_iqm": 0.46979370481231647
  }
},
"forgetting_P": 0.575,
"forgetting_min": 0.25,
"rule": "decide_promotion(G2, G1 on the 9-task battery; G1 = 0 on new tasks) and P(G2 > G1 on G1 tasks) >= forgetting_mi",
},
"lineage_ledger": {
  "path": "/media/knight2/EDS2/toddler-generations/lineage.jsonl",
  "verified": true,
  "code": {
    "commit": "9e4dfc29a88b2ed215232c31555a54ad9577c4a6",
    "dirty": false,
    "remote": "git@github.com:virtuanalytica/toddler.git"
  }
}
}
}

```

10. Afgeleide agents en transmorphoses

agent	soort	doel	goedgekeurd door
-	-	nog geen agents op deze generatie gebouwd	-

11. Integriteit

toddler	sha256 gewichten (nu gemeten)	register + meta.json
G2/t2001	129643dcbbfb35ac752eeb3bbf0a3858a08d107ca1914a353510265d2c5eec1f	ok
G2/t2002	6aba323367f8a0967f9028e1c51fabef584808ca08f31f0f04685b915c5accca	ok
G2/t2003	16ec22df7a650c0b720a7f5125b794213ffed73aa1cba6616d193a679a6a6087	ok
G2/t2004	dfde4eec4229e5e954567e8daecbb5830a18d5e8fe71d9022b001767d05d327b	ok
G2/t2005	811164b29af7821321e44c561d15063ac5855e5d2766ad47d3bec902b76521cc	ok

Hashketen van het evolutieregister: intact; 18 rijen; laatste rijhash df5d09971c4c77a106d1a3add009fe6805ce2165074556118e20eab6df3c5e2e.